

Second Assignment

B.A/B.Sc.(Maths /Geology)/B.A/B.Sc.-B.Ed. (Intg.) (Hons.+ Hons. WithResearch) VI Semester

Introduction to Numerical Methods MATH303

Part 1: Numerical Integration (Gauss Quadrature)

Q1. Two-Point Formula Application

Evaluate the integral $I = \int_1^2 \frac{1}{x} dx$ using the **Gauss 2-point quadrature formula**.

- Note: You must first transform the limits from $[1, 2]$ to $[-1, 1]$.
- Compare your result with the exact value ($\ln 2$).

Q2. Three-Point Precision

Use the **Gauss 3-point formula** to approximate $\int_0^1 e^x dx$.

Part 2: Divided Difference Interpolation

Q3. Polynomial Construction

Using **Newton's Divided Difference Formula**, find the interpolating polynomial for the following data set. Then, estimate the value of $f(3)$.

x	0	1	2	4	5
$f(x)$	1	14	15	5	6

Q4. Higher-Order Estimation

Determine $f(8)$ using divided differences for the following data:

(4, 48), (5, 100), (7, 294), (10, 900), (11, 1210).

Part 3: Central Difference Interpolation (Stirling & Bessel)

Note: These formulae are best used when the point of interpolation is near the center of the data set.

Q5. Stirling's Formula

Given the following table, use **Stirling's Formula** to find the value of y at $x = 1.22$.

x	1.0	1.1	1.2	1.3	1.4
y	0.841	0.891	0.932	0.963	0.985

Q6. Bessel's Formula

Use **Bessel's Interpolation Formula** to find $f(25)$ from the data provided below:

x	20	24	28	32
$f(x)$	2854	3162	3544	3992

Q7. Comparative Analysis

Explain the specific conditions under which you would choose **Bessel's formula** over **Stirling's formula**. (Hint: Consider the value of p in the interval $[0, 1]$).

Assignment Guidelines:

1. **Tabulation:** For all interpolation problems, construct the full difference table (Divided Difference or Central Difference) before applying the formula.
2. **Significant Figures:** Carry out all calculations to at least four decimal places.
3. **Verification:** For Q3 and Q4, verify that your resulting polynomial passes through at least

one of the original data points.